

Effect on Poly(vinyl alcohol) concentration in the transmittance and sheet resistance of conductive PEDOT:PSS/PVA electrodes made by doctor blading deposition technique

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ABSTRACT. The conductive polymer poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS) was adhered onto Poly(vinyl alcohol) (PVA) substrates by the doctor-blading deposition technique. PVA polymer was prepared in an aqueous solution at different concentrations in order to analyze their effect on the manufacture of each flexible conductive film under ambient conditions and their optoelectronic properties. The characterization by Infrared Spectroscopy detected the bond vibrations of the material covering the top of PEDOT:PSS/PVA films (Electrode). Nonetheless, this characterization was complemented with Raman spectroscopy to corroborate the individual characteristic of each material on top of the electrode. In order to explain in more detail the interaction of PEDOT:PSS and PVA during the coating by this

technique, computational modeling was performed using a Reactive Force Field. The results determine the anchoring among PEDOT:PSS and PVA, which is given by the Oxygen and Sulphur bridges coming from the hydroxyl radicals (-OH) and S-thiophene between PVA and PEDOT:PSS, respectively. Additionally, the strong interaction of the conductive side of PEDOT:PSS (PEDOT molecule) and PVA could cause the PSS dissociation from PEDOT, leaving the possibility of PSS to interact with PVA. When, this occurs, both polymers fold into knotted topologies, which could create a bulge on the electrode surface during the coating process. This could explain the increase of the sheet resistance on the conductive film. Likewise, the optical property of the best electrode made by different concentrations of PVA and equal concentration of PEDOT:PSS showed a $T \% = 73 \%$ of transmittance for a $\lambda \approx 457$ nm in the visible spectrum, and a minimum of sheet resistance ≈ 5.84 K Ω /square. Moreover, the topography analyzes of these electrodes shows that the PVA film are covered by PEDOT:PSS and further formation of terraces whatever the concentration used of PVA. Whereas, the morphology and elemental composition properties of the flexible electrodes characterized by scanning electron microscopy (FE-SEM) and Energy-dispersive X-ray spectroscopy (EDS) respectively identified sulfur in the atomic elemental composition of the films with less sheet resistance as well as homogeneous thickness of PEDOT:PSS onto PVA. Therefore, this work presents a detailed study of PEDOT:PSS/PVA electrode with the main goal of producing a plain conductive substrate with an available technique.

INTRODUCTION

The n-type and p-type solid state organic conductors and semiconductors have revolutionized the field of electronic devices industry (eg, transistors, electrical resistors, LEDs, solar cells, etc.) due to their stability, durability, performance, and low cost. These over the years were taking a different

approach in their sustainability and in the improvements of their mechanical properties, for example ITO, ZnO: B, ZnO: Al and SnO₂¹ have been used as electrodes or counter electrodes to transport electrons (e-) and holes (h+) due to its good morphological, optical and electrical properties.

On the other hand, the advance in the field of eco-friendly electronic technologies has proposed new challenges in the state of the art of organic semiconductors,² from simply synthesizing polymers with electrical properties^{3, 4} to the optimization of its optical⁵ and mechanical^{6, 7} properties. It allows the creation of new wearable optoelectronic devices with these properties, achieving notable advances in the field of flexible organic electronic devices^{8, 10} (smart displays, OLEDs, super capacitors, batteries, etc.), becoming an alternative to synthetic metal oxide electrodes that possess properties like semiconductors and conductors.

One of the most used materials in the optoelectronic industry has been Poly (3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS), an n-type conductive polymer with high expectations for manufacturing transparent and flexible electrodes with the lowest resistivity for applications in solar cells,^{11, 12} OLEDs,^{13, 14, 15} sensors^{16, 17} and so forth. PEDOT:PSS turns out to be ideal for the manufacture of conductive thin films due to its good electrical (high conductivity ca. 10 Sxcm⁻¹), optical, mechanical,^{18, 19} thermal (no change in conductivity after treating in the air at 100 °C for 1000 h.)^{20, 21} and elasticity (Young's modulus)²² characteristics. This material has been deposited on different kind of polymer matrix such as PET and PVA. In this context, the use of PVA represents an ecofriendly and easy electrode manufacturing process in aqueous solution with good physical property.^{24, 25} However, PVA holder presents disadvantages due to that its optical properties depends on the concentration in water and it can be degraded in the environment.

Thus, all these topics needs to be studied in more detail in order to consider the correct manufacturing technique through the doctor blading.

In this work, we report the manufacture of a flexible conductive substrate based on PEDOT:PSS supported on PVA (PEDOT:PSS/PVA) through the doctor blade technique. The PEDOT:PSS thin film thickness was measured indirectly from the UV-Vis spectrum by means of the equation of maximum and minimum. This method allows an affordable measurement of coating on flexible polymeric substrate. Moreover, the transmittance and sheet resistance values of PEDOT:PSS/PVA thin film with the best performance reached values $\approx 73\%$ and $\approx 5.84\text{ K}\Omega/\text{square}$ respectively. In addition, the molecular simulations of the binary combination of the polymers allows a better understanding of their interaction into the matrix. Our results showed that the -OH functional groups are responsible for the anchoring between the PEDOT:PSS and the PVA molecule. Furthermore, the role of hydroxyl groups in interchain interactions was corroborated by Infrared and Raman spectrophotometry onto the substrates. While, the use of microscopy of Atomic Force (AFM), Field Emission Scanning Electron (FE-SEM) and Energy-dispersive X-ray (EDS) allows to measure a good dispersion of the sulfur element on the PVA surface, which could explain an homogeneous cover conductive material (PEDOT:PSS).

EXPERIMENTAL

Materials

The chemical reagents such polymer poly(3,4 ethylenedioxythiophene):polystyrene sulfonate (1.3wt% dispersion in H_2O , conductive grade) and Poly(vinyl alcohol) (Mw 85000-124000, 99+% hydrolyzed) were purchased from Sigma Aldrich.

Assembly of electrodes based on aqueous solution by doctor-blading deposition

Solutions with different concentrations of PVA (**Table 1**) were prepared using ultra-pure water and a magnetic stirrer equipped with a magnetic stir bar and a hot plate. These solutions were taken to a Polyethylene terephthalate (PET) mold and dried at 65 °C in an oven. After detaching (peel off) both materials, transparent flexible PVA films were obtained, and they were cut into square sheets with dimensions of 2.5 x 2.5 cm. Once the PVA substrates were manufactured, equal concentrations of PEDOT:PSS (**Table 1**) were filtered and diluted in ultra-pure water in a 10 ml flask equipped with a magnetic stir bar. These solutions obtained were used to coat the PVA substrates through doctor-blading deposition (**Figure 1 a**) controlling the drying process with hot air up to 60 °C.

Table 1. Concentrations of PEDOT:PSS and PVA diluted in ultra-pure water.

PEDOT:PSS/PVA electrodes	PEDOT:PSS (%)	PVA [g/L]	PVA substrate thickness (mm)
1	80	20.212	1
2	80	28.044	1
3	80	32.126	1

Chemical and morphological characterization of PEDOT:PSS/PVA electrodes

In order to detect the frequencies of the different functional groups linked to PVA film, an FTIR spectrophotometer (Thermo Fisher Scientific model Nicolet iS10) in Attenuated total reflectance (ATR) mode was used. This was complemented by Raman spectroscopy characterization using an equipment brand NT-MDT equipped with two lasers of 532 nm and 630 nm wavelength. Followed by FE-SEM characterization using a Field Emission Scanning Electron Microscope - HITACHI model SU8230. Likewise, we characterized the elemental composition of the PEDOT:PSS onto PVA by the X-Ray Energy Dispersive Spectroscopy (EDS) using a Bruker multichannel detector model Xflash FladQUAD 5060F. The AFM images were generated using an SPM/SNOM/TERS

NTEGRA SPECTRA II equipment of brand NT-MDT and each image was analyzed with Gwyddion 2.53.

Optical and electrical characterization of PEDOT:PSS/PVA electrodes

The electrodes transmittance was measured using a UV-Vis spectrophotometer model USB 4000 brand Ocean Optics, equipped with two lamps (a deuterium and a halogen DH-200-BAL brand - Ocean Optics), which produce electromagnetic waves ranging from 210 nm to 800 nm. The PEDOT:PSS onto PVA sheet resistance measurement was carried out by four-pointed equipment (**Figure 1 b**). This was equipped with Keithley 6220 equipment to measure the current and Keithley 2701 equipment with a modified 7709 card to measure the voltage and silver paint was used to improve the contacts between the points and the film-coated as well.

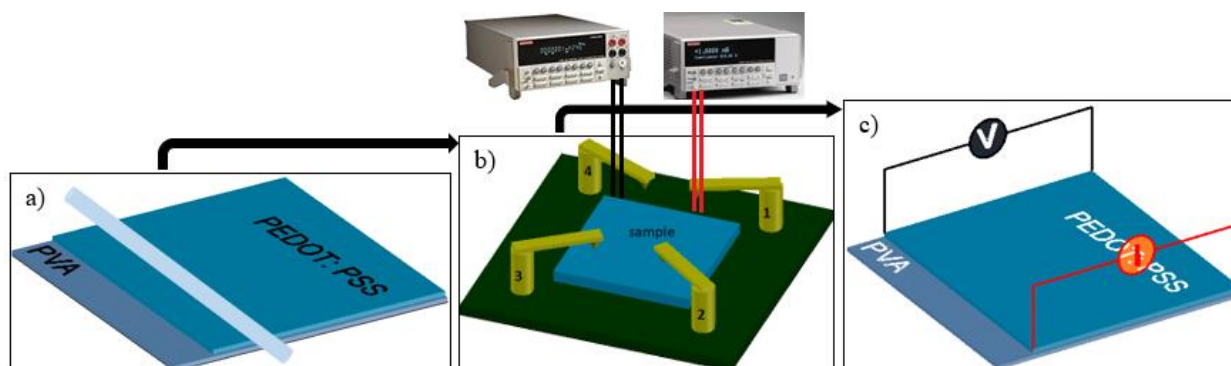


Figure 1. Process schematic to manufacture of PEDOT:PSS/PVA electrodes (a), four-point device (b) and its output configuration of voltage (V) and input of current (I) (c).

Computational Model

The molecular models for our tested polymers were based on the chemical structure of the monomer of each component (PEDOT:PSS and PVA) at dry condition and neutral pH. We simulated four binary systems, which consists in the combination of PEDOT-8 monomers, PSS-8

monomers and PVA-32 monomers. The PEDOT:PSS system was in stoichiometry ratio at pH 7.0, where the PSS polymer has had partially protonated. To balance the charge of the entire complex, we added some sodium atoms as counterions, which are present in the commercial sample. The PVA polymer has been considered protonated because its pKa is 10.67. The molecular dynamic simulations were performed using the Reactive Force Field, which is implemented in LAMMPS software²⁷. Moreover, the parameter C/H/O/all_carbon/sulfonate/phosphate/Ti/Na/Cl/water²⁸ was used for these simulations because it was previously tested for organic polymers as peptides. In addition, Qeq was used to equilibrate the charges according to the electronegativity of the atoms. Afterward, the molecular systems were equilibrated using the NVT ensemble after the NVE ensemble. The structures were equilibrated at constant temperature of 300° K with the 0.25 fs time step. The initial structure for the evaluated interfaces were PEDOT–8 monomers/PEDOT–8 monomers, PVA–32 monomers/PEDOT–8 monomers, and PSSH-PSSNa–4 monomers. Each component was separated by 6.0 Å and the counterion is separated from other atoms at 3 Å.

RESULTS AND DISCUSSION

The relative transmission spectra of PEDOT:PSS/PVA electrodes in the MIR region between 800 cm⁻¹ and 4000 cm⁻¹ are shown in **Figure 2 a**. There characteristic vibrational frequencies of various chemical species were detected, the PVA spectrum (line color cyan) shows regions with absorption bands at 840 cm⁻¹ ν ~(C–H), 918 cm⁻¹ ν ~(H–C=CH₂) around 1091 cm⁻¹ ν ~(C–O–C), 1336 cm⁻¹ ν ~(C–O), 1420 cm⁻¹ (CH₂), 1563 cm⁻¹ ν ~(C=C), 1653 cm⁻¹ (C=O), 2911 cm⁻¹ – 2946 cm⁻¹ ν ~(C–CH₃) and a width band from 3008 cm⁻¹ to 3667 cm⁻¹.^{29, 30} The last very broad band can be assigned to ν ~(O–H) vibrations and arises due to the presence of carboxyl and hydroxyl groups. This absorption band at the visible region does not show a clear interaction due to the strongly varying strength of the hydrogen bonding of these functional groups. **On the other hand,**

PEDOT:PSS (line color purple) shows weak signals in the region where their relevant functional groups are i.e. C–S thiophene and S–phenyl²⁹. These were detected weakly in the absorption band at $831\text{ cm}^{-1} - 921\text{ cm}^{-1}$ the $\nu\sim$ C–S thiophene, at 1087 cm^{-1} (characteristic $\nu\sim$ S–Phenyl), and SO_3 (asymmetric stretch) at 1279 cm^{-1} . These bands overlap between them when PEDOT:PSS is deposited on the PVA polymeric molecules.³¹ According to this, after PEDOT:PSS deposition onto PVA flexible substrates by doctor-blading technique the analyzed electrodes (1, 2 y 3) by IR have shown sharp bands at the same wavenumber as the PVA. This issue has not been reported with detail in previous studies. However, the possible adherence for either PEDOT or PSS molecule at the moment of being spread onto the PVA polymer is stabilized by interaction by OH functional groups, which is revealed by Raman spectroscopy and computational molecular model.

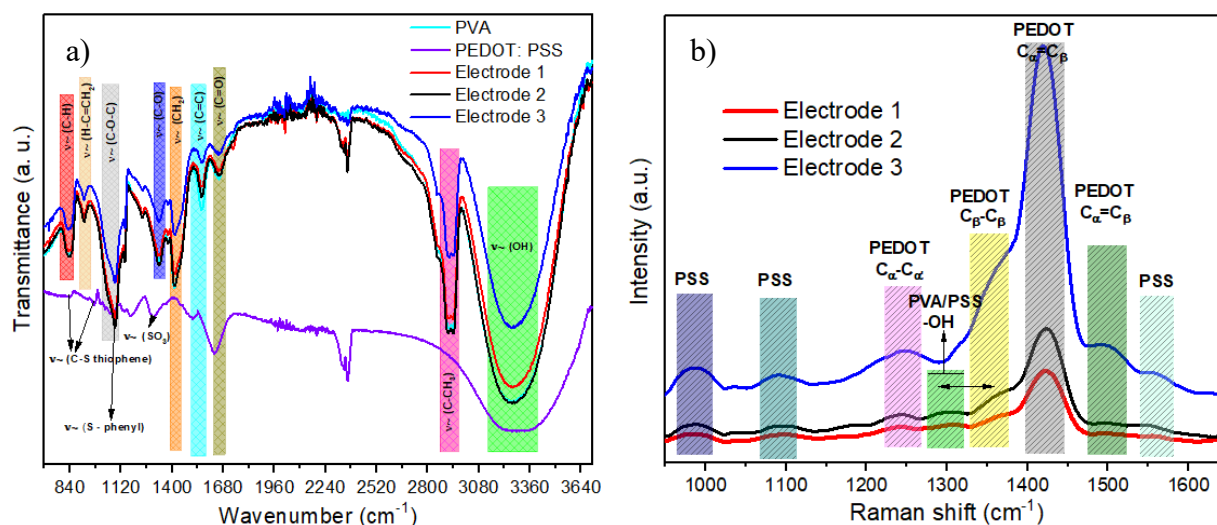


Figure 2. a) Relative infrared transmission spectra of PVA and PEDOT:PSS films. The spectra are shifted with respect to each other along the y-axis. Characteristic vibrational frequencies of various chemical species are assigned to the observed absorption bands in each electrode of PEDOT:PSS/PVA and Raman spectra of the three electrodes are shown in b).

Raman spectroscopy is a sensitive technique to investigate conformational changes on the PEDOT:PSS films. **Figure 2b** presents the Raman spectrum of electrodes composed by

PEDOT:PSS over PVA and whose peaks of the chain conformation bands of PEDOT:PSS are identified as follows: oxyethylene ring deformation at 990 cm^{-1} , $C_{\alpha}-C_{\alpha'}$ inter-ring stretching at 1255 cm^{-1} , the $C_{\beta}-C_{\beta}$ stretching mode at 1368 cm^{-1} , $C_{\alpha}=C_{\beta}$ symmetric stretching vibrations at 1430 cm^{-1} , $C_{\alpha}=C_{\beta}$ asymmetrical stretching at 1505 cm^{-1} , and the $C=C$ asymmetric stretching vibration at 1568 cm^{-1} .^{32, 33} The vibrational modes of the Raman spectra of the electrode 3 is far different from the corresponding ones from electrodes 1 and 2. The change is mainly observed between the peaks of the $C_{\alpha}-C_{\alpha'}$ interring stretching, and the $C_{\beta}-C_{\beta}$ stretching mode, which occurs between 1300 and 1353 cm^{-1} . This vibrational mode is associated with $-\text{OH}$ functional group of PVA (Raman shift 1310 cm^{-1}) and is not very intense.³⁴ Therefore, this favorable interaction between the components emphasizes the electronic conductivity that exists between the PEDOT:PSS and PVA molecules, which due to the higher concentration of the PVA support polymeric matrix makes this vibration attenuated while the $C=C$ vibrational mode is intensified at electrode 3. The results of molecular dynamics simulation exhibit the stability of PEDOT-8 dimer at 300 K because of the $\pi-\pi$ interaction, and the $\text{S}-\text{O}$ interaction, which are strong interactions (higher than our other binary models). Studies have shown that PEDOT:PSS film forms a granular behavior, in which the core is composed of the stacked PEDOT layer surrounded by the PSS polymer. This scenario occurs at electrode 1 when the PVA concentration is not enough to disrupt the strong PEDOT – PEDOT interaction. This behavior is shown in **Figure 7**, where the accumulation of the C element is enclosed by the S element region corresponds to the PEDOT clusters rich area and PSS rich area, respectively. Hereinafter, has been observed in **Figure 3** that when PEDOT forms a cluster, the distance $C_{\alpha}-C_{\beta}$ L1 ($C_{\alpha}-C_{\beta}$ L2) bond becomes longer in comparison with the monomer $1.359\text{ \AA}$ (suggesting a quinone-PEDOT like structure). It creates a resonance in the thiophene ring, facilitating charge transport on the surface.

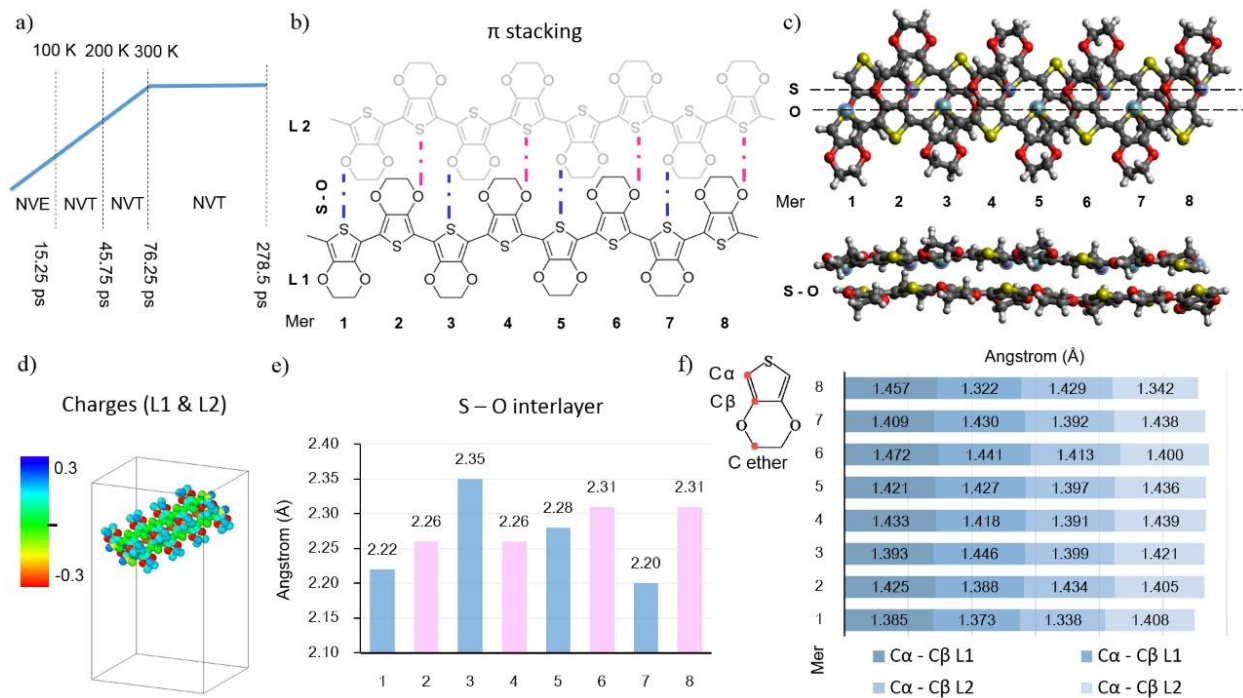


Figure 3. Structural analysis of the binary system PEDOT 8/PEDOT 8 after the NVE and NVT equilibration at the 300° K (278.5 ps). A) Temperature ramp for the MD simulation. B) Schematic contact map between PEDOT layers, top layer is L1 and down layer is L2, colored lines shows S–O interaction. C) Top and side view of the equilibrated PEDOT dimer at the 300 K composed by 8 monomers (Mer.). d) Qeq charges of the molecular system and simulation box. E) Distance of the S–O between PEDOT layers in Angstrom. F) Distance of the $C_{\alpha}=C_{\beta}$ bond at the thiophene ring in Angstrom, where L1 means top layer and L2 down layer.

It facilitates the formation of PEDOT that allows crystallization, its simulations show π – π stacking of the PEDOT crystalline are close enough to arrange themselves at the distance between layers around 2.28 Å. On the other hand, the interaction between PEDOT and PSS layer, **Figure 4** shows that the PSS polymer is flexible around the PEDOT, causing a little twist along the central ax. It is also noticed that two oxygen atoms from the PSS doped the S–PEDOT at two locations (magenta lines). This indicates the presence of polarons in the conductive area. In addition, there are two

hydrogen bonds between the protonated sulfonate of PSS and the hydrogen at the C ether of PEDOT (blue lines) according to Raman bands (**Figure 2**).

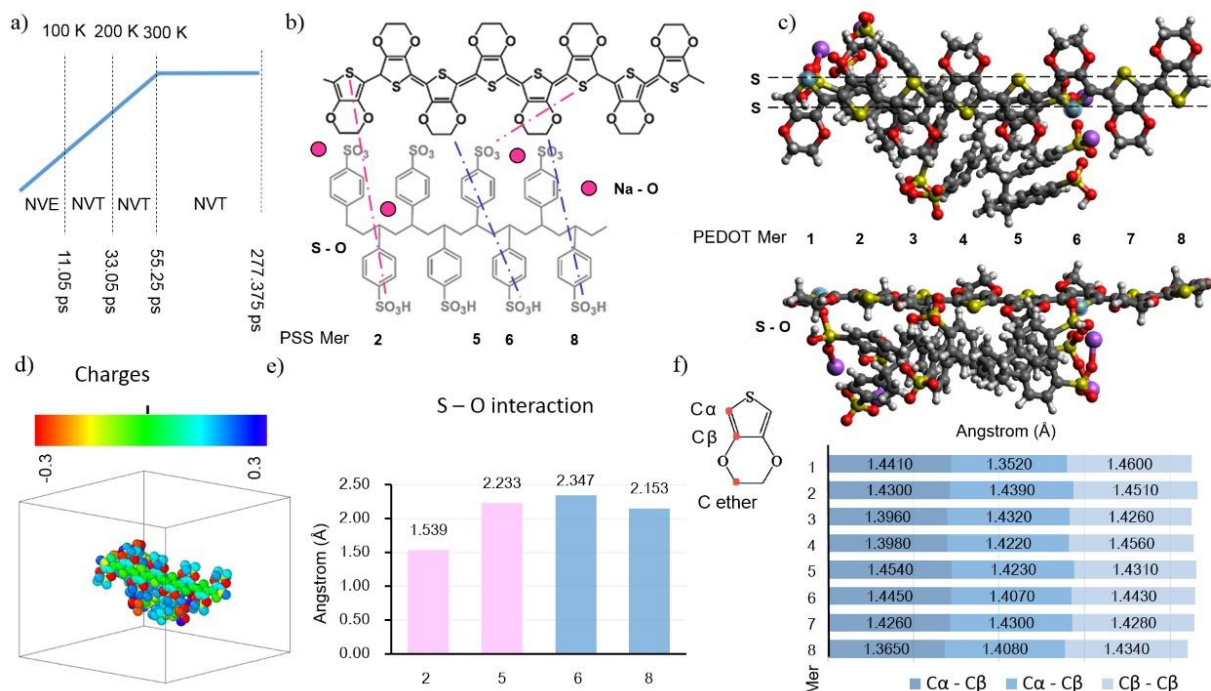


Figure 4. Structural analysis of the binary system PEDOT – 8/partially protonated PSS – 8 with sodium as counterions after the NVE and NVT equilibration at the 300 K (277.375 ps). A) Temperature ramp for the MD simulation. B) Schematic contact map between PEDOT and PSS layers, colored lines show different type of interaction: magenta means S–O and blue H–O interaction, magenta circle shows the presence of sodium atoms at the interface. C) Top and side view of the equilibrated PEDOT / partially protonated PSS at the 300° K composed by 8 monomers (Mon.). d) Qeq charges of the molecular system and simulation box. e) Distance of the S–O between PEDOT and the partially protonated PSS layer in Angstrom. f) Distance of the C α =C β and C β –C β bond at the thiophene ring in Angstrom.

In the same way, **Figure 4** shows a strong interaction between sulfonate groups with sodium atoms, each sodium atom binds two sulfonate groups. This interaction reconfigures the morphology of

the PSS polymer, generating loops at the interface. On the other hand, the S–O interaction is strong, especially between the first monomer of PEDOT and the second monomer of the PSS (from the left to the right), which has 1.539 Å. Further, PEDOT disperses over the PVA layer thanks to the following 6 contacts among them, which is S–O (OH) type of interaction. This interaction causes the OH band of the PVA to become broad in the Raman spectra (**Figure 2**). Through this event, PVA stabilizes the PEDOT at the horizontal accommodation (**Figure 5**) to interact with either PEDOT or PSS layers.

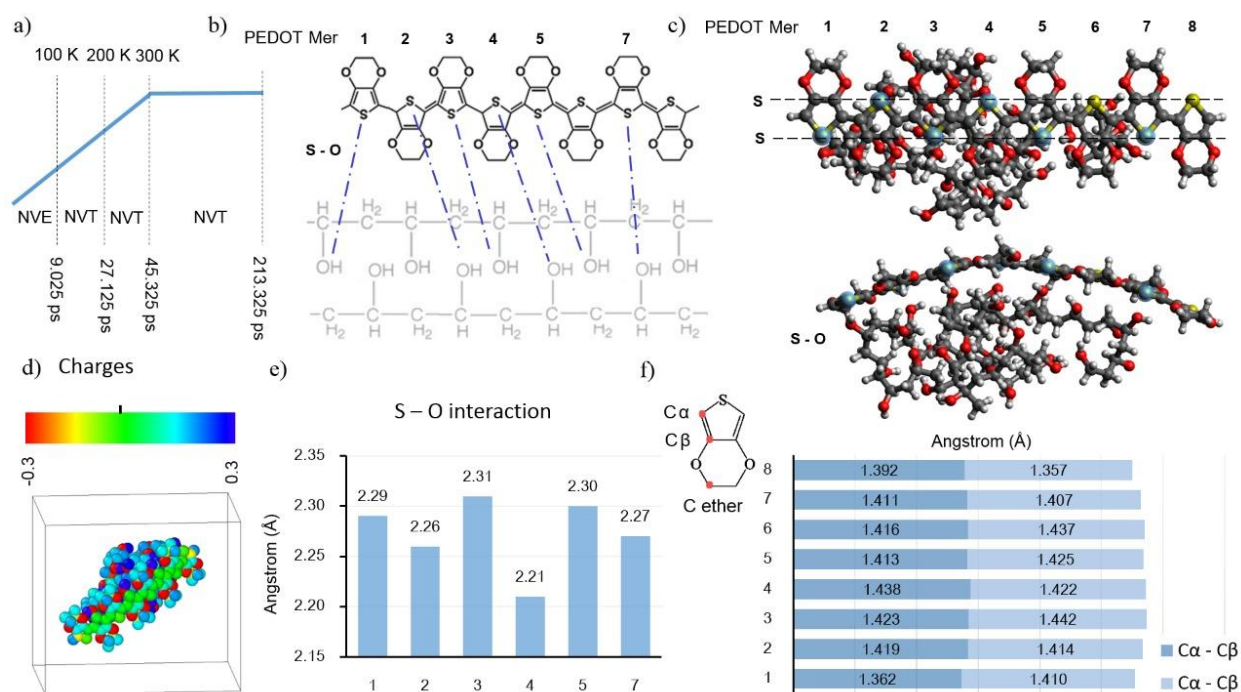


Figure 5. Structural analysis of the binary system PEDOT–8/PVA–32 after the NVE and NVT equilibration at the 300 K (213.325 ps). a) Temperature ramp for the MD simulation. b) Schematic contact map between PEDOT and PVA layers, blue lines show the S–O and interaction. c) Top and side view of the equilibrated PEDOT/PVA at the 300 K composed by 8 monomers (Mer). d) Qeq charges of the molecular system and simulation box. e) Distance of the S–O between PEDOT and PVA layer in Angstrom. f) Distance of the Cα – Cβ at the thiophene ring in Angstrom.

According to the experiment, whether the other components are present, the organization between them will occur to facilitate the flux of current. Simulations suggest that PVA polymer fix PEDOT molecules better than PSS and creates a homogeneous distribution, and experimentally we observe that if the PVA is presented in a low concentration, PEDOT could form clusters immersed into the PSS network (Electrode 1) unable to easily interact with other components, leaving a part of PSS to form knots-like topology. This could increase the roughness on the surface of PEDOT:PSS and PVA film. In consequence, the sheet resistance of electrode increase up as occurs in electrode 1. Indeed, PEDOT shows good dispersion over the PVA surface (Figure 5), even though the contact with PSS is less spatial distributed (Figure 6).

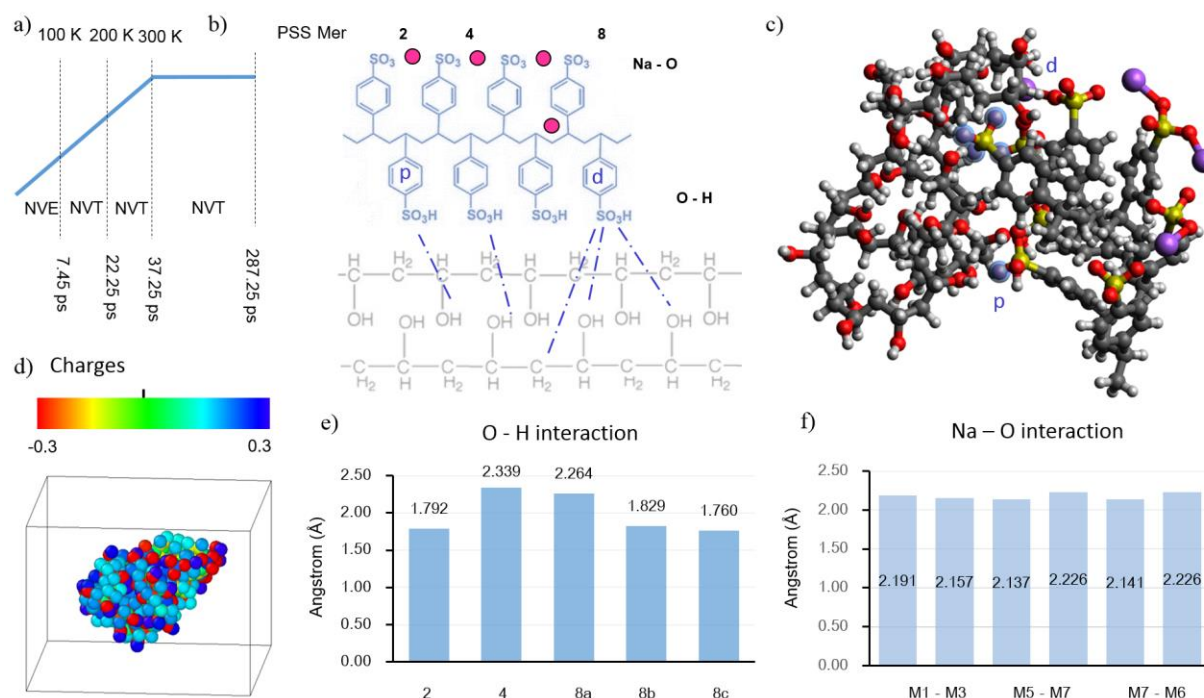


Figure 6. Structural analysis of the binary system of partially protonated PSS - 8/PVA - 32 with sodium atoms as counterion after the NVE and NVT equilibration at the 300 K (287.25 ps). a) Temperature ramp for the MD simulation. b) Schematic contact map between partially protonated PSS and PVA layers, blue lines show the O-H interaction, the letter “p” means protonated and “d”

deprotonated state. c) Side view of the equilibrated PSS/PVA at the 300° K composed by 8 monomers (Mer) and 32 monomers. d) Qeq charges of the molecular system and simulation box. e) Distance of the O–H interaction at the interface of partially protonated PSS and PVA in Angstrom. f) Distance of the Na–O interaction between the sulfonate from PSS polymer and PVA in Angstrom. M1, M3, M5 M6 and M7 mean the order of the monomer from the left to the right.

In the PSS–PVA system, the interface interaction is composed mainly of hydrogen bonds, which are less energetic than S–O interaction. Thus, this system is not energetically favorable in comparison with PEDOT–PVA interaction. Interesting, when PVA interacts with PSS proton transfer occurs, which could create less efficient charge transfer into the electrode conductive film. According to the AFM images (Figure 8), when PVA concentration increases, the film topography changes, which means the roughness decrease. Hence it does not exclude PVA interacting with the outer PSS shell as well and it generates a phase due to the poor dispersion among these materials, which would cause the rugosity at the grain barrier as we observe in electrodes 1 and 2 (Figure 7). Besides, PSS under the contact of PVA becomes very flexible and the close contact among monomers causes an intra-protonation state between sulfonates at the interface. The protonation at the interface could cause morphological changes in the PSS. For instance, the anionic sulfonate from PSS has three S–O interactions with PVA, while protonated sulfonate has only one.

In the same way, these results are related to the morphology of the PEDOT:PSS surfaces. The FE-SEM images (Figure 7) demonstrate the surface characteristic of the PVA substrate coated with PEDOT: PSS. The three electrodes were subjected to this characterization, finding the coating defects on the PVA support substrate by the doctor blading deposition technique. Electrode number 1 exhibits defects on the conductive side like folds and lumps due to agglomeration of the

PEDOT: PSS in greater proportion in different areas than others. This is verified through the EDS images were S atoms linked to the C–S thiophene, and S–Phenyl species are found in the coating surface of the electrode in an irregular way. The close to the uniform coating of these chemical species characteristic of PEDOT:PSS onto PVA could be observed by maintaining the concentration of PEDOT:PSS at 80 % and increasing the PVA concentration in the solution to values of 28.044 % and 32.126 %. In this sense, electrode number 2 showed small lumps with an average radius of 0.022 μm . This is attributed to the PEDOT since its EDS image shows that the terrace-shaped areas have an elemental composition of C and O being surrounded by the S element stemming from the PSS.²⁹

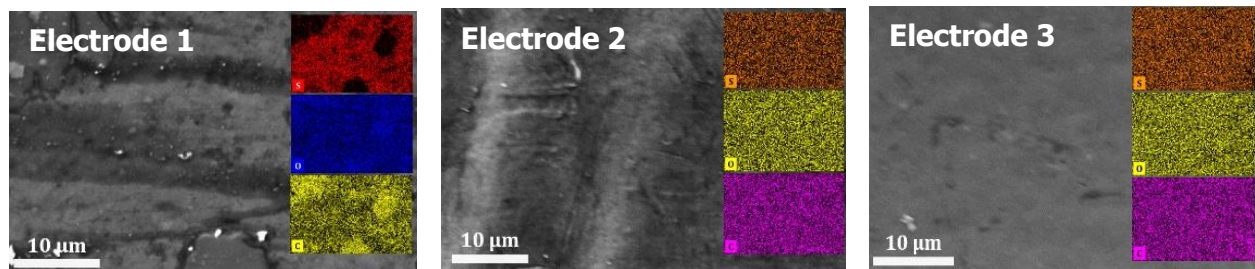


Figure 7. FE-SEM images with respective EDS analysis images of PEDOT: PSS/PVA electrodes manufactured by doctor blading deposition technique. In Electrode 1 Sulphur (red color), Oxygen (blue color), and Carbon (yellow color), while in Electrodes 2 and 3 Sulphur (orange color), Oxygen (yellow color), and Carbon (purple color).

In contrast to this, in electrode number 3 it can be seen the formation of lumps has decreased, reaching a surface with a more uniform coating. Therefore, the defects on the conductive side of the electrode 2 and 3 decrease in order to increase the PVA concentration and keeping constant the PEDOT:PSS concentration.

These three electrodes were analyzed by AFM (**Figure 8**). The arithmetic means roughness (R_a) of the electrodes was analyzed according to **Equation 1**. Moreover, it can be seen in the 3D images

of the electrodes (**Figure 8**) that the roughness along the topography varies according to the increase of PVA concentration making the dispersion of PEDOT:PSS on the holder becomes more homogeneous. Further, **Figure 8 a, b and c** show the direction of the displacement of the PEDOT:PSS solution on the PVA. Appearing flat terraces one after the other, which were formed when spreading the PEDOT:PSS drop with the glass roller on the PVA film.

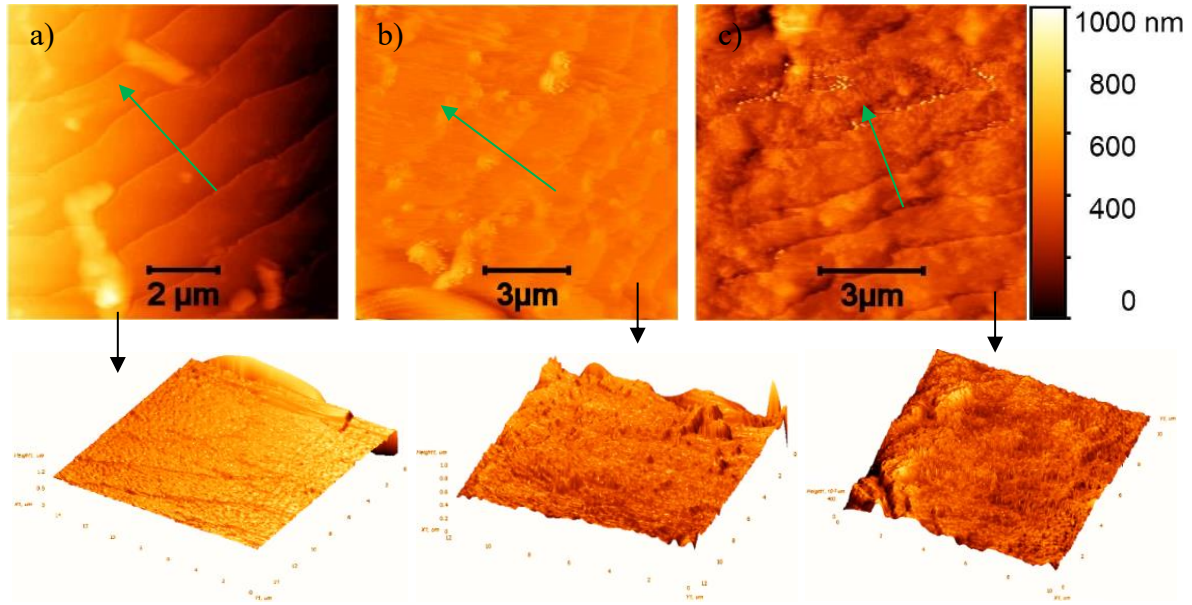


Figure 84. AFM images to analyze the topography of PEDOT: PSS/PVA electrodes fabricated by the doctor blading deposition technique, a) electrode number 1, b) number 2 and c) number 3. The green arrow shows the direction of coating with glass roller.

The formation of these terraces was also influenced by the temperature, in which they were dried during the coating process.

$$R_a = \frac{1}{L_m} \int_0^{L_m} |y| dx \dots \text{Equation 1}$$

Where L_m is the measured length, a is the height and y is the ordinate. Likewise, the calculation of the R_a was corroborated by **Equation 2** using Gwyddion software.

$$R_a = \frac{1}{n} \sum_{i=1}^n |y| \dots \text{Equation 2}$$

Where n is the number of ordinates considered. According to this analysis the conductive side of electrode number 3 has a $R_a = 3.64$ nm, this is due to PVA concentration in the film thus improving its mechanical properties, thereby resisting the coating of the PEDOT:PSS solution on its surface. Furthermore, this interaction between PEDOT:PSS and PVA causes the optical properties of the conductive side to change. In **Figure 9 a**, electrode number 3 shows a maximum transmittance (T_M) in the visible spectrum reaching a value ≈ 73 % for $\lambda \approx 457$ nm. Similarly, for electrode number 2 with a $R_a = 6.338$ nm its T_M reaches a value ≈ 60.91 % for $\lambda \approx 593.45$ nm and for electrode number 1 whose $R_a = 9.853$ nm the T_M is ≈ 69.85 % for $\lambda \geq 765$ nm. Corroborating with this, the increase of roughness in each film due to the formation of PSS agglomeration on the surface of PEDOT and PVA. As well, **Figure 9 a** shows that electrodes number 2 and 3 have T_M and minimum transmittances (T_m), which are directly related to the thickness of the PEDOT:PSS film and the PVA substrate.³⁵

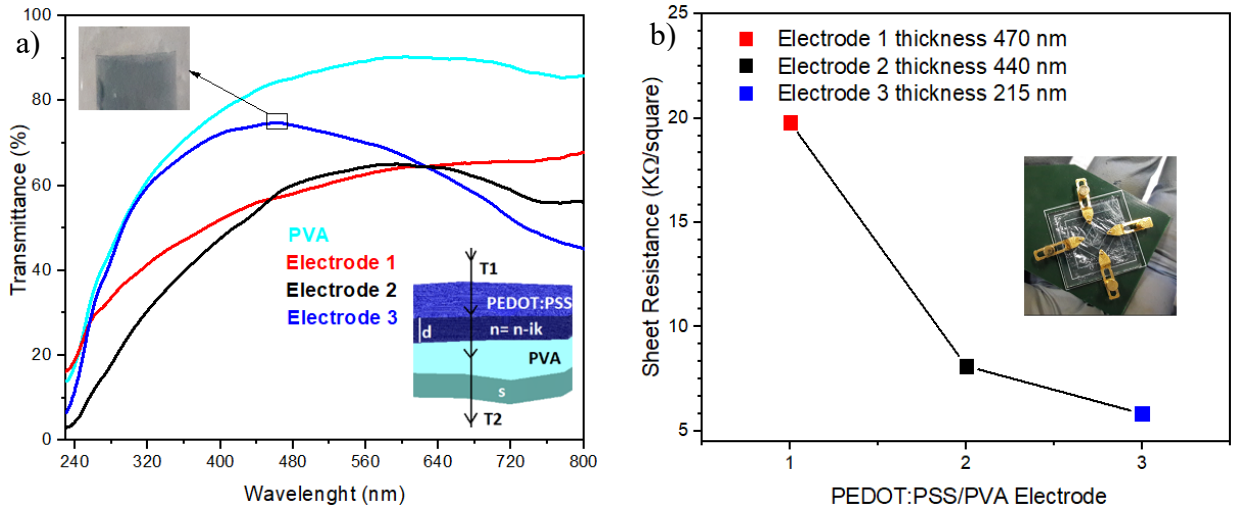


Figure 9. a) T_M and T_m of PEDOT:PSS/PVA electrodes fabricated by the doctor blading technique. b) Change in the sheet resistance (blue squares) according to the film thickness of the PEDOT:PSS on PVA.

The smooth curves shown are due to the fact that the thickness of the PEDOT:PSS film onto PVA substrate is not uniform, destroying all interference effects. However, fringes can be seen on electrodes 2 and 3 caused to the lower agglomeration rate of PEDOT:PSS on the support substrate. While for electrode 1 it is the opposite, which tends to increase its transmittance as the wavelength passing through it increases. These fringes have also been observed in thin films formed on solid substrates and whose equations can be used to calculate the optical constants³⁵ on the conductive side of the electrodes showing T_M and T_m . The basic mathematical representation of the fringes is given by **Equation 3**:

$$2nd = m\lambda \dots \text{Equation 3}$$

Where n is the refractive index, d is the PDOT:PSS thin film thickness and m is an integer for maxima and half integer for minima. While the extremes of the interference fringes are represented by **Equations 4 and 5**:

$$T_M = \frac{Ax}{B - Cx + Dx^2} \dots \text{Equation 4}$$

$$T_m = \frac{Ax}{B + Cx + Dx^2} \dots \text{Equation 5}$$

Where:

$$A = 16n^2s$$

$$B = (n + 1)^3(n + s^2)$$

$$C = 2(n^2 - 1)(n^2 - s^2)$$

$$D = (n - 1)^3(n - s^2)$$

$$x = \exp(-\alpha d)$$

Where s is index of refraction and α absorption coefficient. **Equation 6** represents this tendency of its transmittance (T).

$$T = \frac{Ax}{B - Cx \cos\varphi + Dx^2} \dots \text{Equation 6}$$

Where:

$$\varphi = \frac{4\pi nd}{\lambda}$$

Based on the results of the average thickness of the PEDOT: PSS film obtained through the AFM analysis and whose values are shown in **Figure 9 b** and by replacing the average value of $n = 1.2 \sim 1.6^{36, 37}$ in **Equation 3**, calculated the value of the coefficient m for electrodes 2 and 3 using their maximum transmittances in the region of strong and medium absorption respectively, obtaining as a result 2.31 for electrode number 2 and while for electrode 3 a value of 0.87.

The measured resistance of the PEDOT:PSS films (**Figure 5 b**) shows the electrical performance of electrodes according to the variation of their thickness and PVA concentration in the matrix substrate. According to the graph, electrode number 1 presents a high sheet resistance (19.79 K Ω /square) while electrodes 2 and 3 present a low resistance of 8.11 and 5. 84 K Ω /square to the passage of the current respectively, that due to the increase of the concentration of PVA, which causes the greater number of moles of PVA to affect the rearrangement of the PEDOT and PSS on the surface of the electrode, interfering with the passage of the current through the conductive polymer acting also on the anisotropic properties of each material at the moment of the passage of the flow of electrons.

CONCLUSIONS

In conclusion, the 1 mm thick PVA films coated with PEDOT:PSS by the doctor blading technique showed desirables optical and electrical properties upon to the concentration variation in their polymeric matrix. Computational chemical modeling identified different types of interactions between PEDOT and PSS molecules to PVA. This chemical interaction occurs between –OH and S–O functional groups, and affect the distribution of the layers. Higher PVA presence could favor

the PVA and PEDOT interaction because to its higher interaction rather than PVA and PSS, which is strongly linked to the enhancement of the conductive film on the substrate. Additionally, the concentration of PVA favors the interaction between PVA and PEDOT, which is energetically favorable in comparison with the other heterogeneous binary system PEDOT – PSS and PVA – PSS. In that sense, the concentration in the PVA film plays a very important role in the optical and electrical structural performance of the conductive electrode through its maximum and minimum transmittance, which allows us to know that these properties are not only present in n-type semiconducting electrodes made of metal oxides, but also in flexible electrodes based on conductive polymers. Moreover, their electronic properties analyzed by the sheet resistance of the 3 electrodes showed that the current flow in PEDOT:PSS is related to the PVA concentration. Therefore, the interaction of the PEDOT:PSS and PVA molecules previously processed in solution plays an important role in the quality of the flexible electrode.

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